



Customer Retention and Churn Risk Analytics

INTERNSHIP PROJECT – 01

Data Analytics Project

 **Internship Organization :** Persevex

 **Intern :** Pratik Patil

 **Department :** Computer Engineering

 **Internship Duration :** 3 Months

 **Project :** Project 01 – ChurnWatch

 **Technologies Used :**  Excel |  SQL |  Power Query |  DAX |  Power BI

Project Focus

Customer Churn Analysis • Risk Scoring • Revenue Risk • Retention Operations

Project Workflow

Data → Cleaning → SQL Analysis → Risk Scoring → Power BI → Insights → Retention

 **INTERNSHIP PROJECT DOCUMENTATION**

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1. PROJECT OVERVIEW

ChurnWatch is a customer churn and retention analytics project developed using Excel, SQL, Power Query, DAX, and Power BI.

The project focuses on analyzing customer behavior, engagement, subscription information, revenue, churn status, and customer risk.

The main purpose of ChurnWatch is to transform raw customer data into meaningful business insights that can help organizations understand customer churn and identify customers who may require retention attention.

The solution follows a complete analytics workflow:

 **Data** →  **Cleaning** →  **SQL Analysis** →  **Risk Scoring** →  **Power BI** → 
Insights →  **Retention Actions**

The project contains four major Power BI dashboard pages:

Executive Dashboard

Provides an overall summary of customers, churn, revenue, and customer health.

Churn Analysis

Analyzes customer churn patterns across different customer characteristics.

Customer Risk Dashboard

Identifies active customers who show potential churn signals and categorizes them into risk bands.

Retention Operations

Provides an operational view of customers requiring retention intervention, including recommended actions, owners, SLA hours, and intervention status.

ChurnWatch therefore combines descriptive analytics, customer-risk analysis, and retention operations into one business intelligence solution.

2. BUSINESS PROBLEM

Customer churn is an important business challenge because losing existing customers can affect recurring revenue, customer relationships, and long-term business growth.

A company may have thousands of customers, but simply knowing the total number of customers does not explain customer health.

Businesses need answers to questions such as:

- ◆ How many customers have churned?
- ◆ What is the overall churn rate?
- ◆ Which customer segments experience higher churn?
- ◆ Are customers becoming less engaged?
- ◆ Which active customers may be at risk?
- ◆ What is the potential revenue impact?
- ◆ Which customers should receive retention attention?
- ◆ What retention actions are currently pending?

Traditional reports often focus on historical results. ChurnWatch extends this analysis by examining customer engagement and risk signals.

The project provides a centralized analytical framework where customer behavior can be analyzed and connected with retention operations.

3. OBJECTIVES

The major objectives of ChurnWatch are:

Objective 1 — Analyze Customer Churn

Measure customer churn and understand overall customer retention performance.

Objective 2 — Calculate Business KPIs

Calculate important metrics such as:

- Total Customers
- Active Customers
- Churned Customers
- Churn Rate
- Revenue
- Revenue at Risk
- Average Revenue per Customer

Objective 3 — Analyze Customer Behavior

Analyze customer engagement using:

- Login activity
- Session activity
- Feature usage
- Usage decline
- Last active date
- Days since last activity

Objective 4 — Identify Customer Risk

Develop a risk-scoring approach to classify active customers into different risk bands.

Objective 5 — Understand Revenue Exposure

Connect customer risk with expected revenue impact.

Objective 6 — Support Retention

Provide recommended actions and operational information for retention teams.

Objective 7 — Build Interactive Dashboards

Develop Power BI dashboards that allow users to explore customer churn and risk using filters and interactive visualizations.

4. 📁 DATASET DESCRIPTION

The ChurnWatch project uses a customer-level dataset containing approximately **1,000 customer records**.

The dataset contains information from multiple analytical areas.

👤 Customer Information

- Customer_ID
- City
- Region
- Customer_Segment

📦 Subscription Information

- Subscription information
- Plan information
- Start date
- Customer tenure
- Subscription status

💰 Revenue Information

- Revenue
- Revenue-related customer metrics
- Expected Revenue Impact

📱 Customer Engagement

- Login_Count_30D
- Sessions_30D
- Feature_Usage
- Usage_Decline_Pct
- Last_Active_Date
- Days_Since_Last_Activity

Churn Information

- Churn_Status
- Churn_Date
- Churn-related attributes

Risk Information

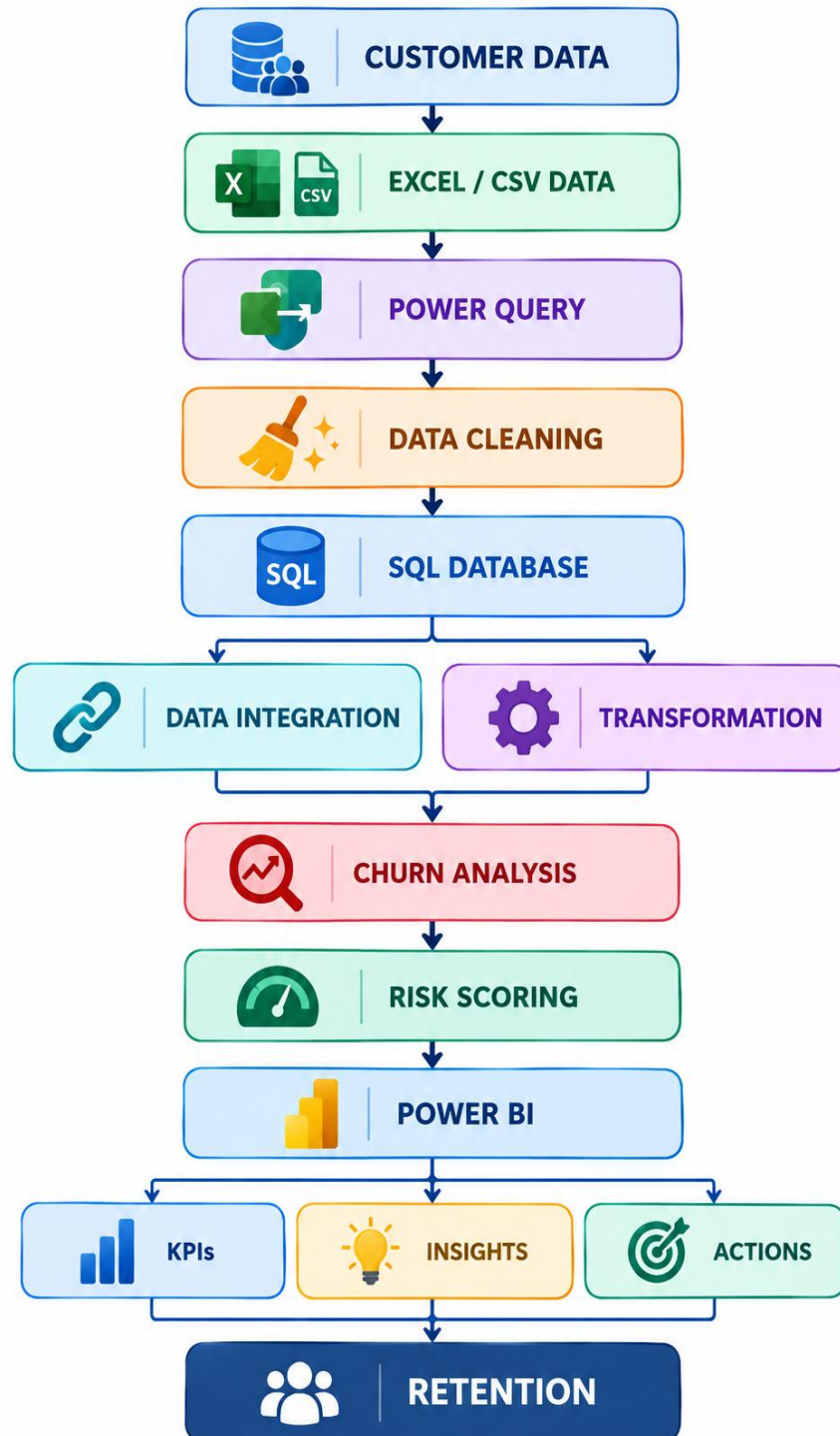
- Risk_Score
- Risk_Band
- Expected_Revenue_Impact_INR

Retention Operations

- Recommended_Action
- Owner
- SLA_Hours
- Intervention_Date
- Intervention_Status
- Intervention_Outcome

5. DATA ARCHITECTURE

The ChurnWatch architecture follows a structured data analytics pipeline.



Main Components

Component	Purpose
 Excel	Initial dataset and data preparation
 Power Query	Data cleaning and transformation
 SQL	Data integration and analytical queries
 Risk Engine	Customer risk calculation
 DAX	Power BI measures and calculations
 Power BI	Interactive dashboard
 Retention Module	Intervention tracking

The Customer_ID acts as the main identifier for connecting customer-related information.

6. 🧹 DATA CLEANING

Data cleaning was performed to improve the quality, consistency, and reliability of the dataset before analysis.

◆ Missing Values

Missing values were identified and handled according to the meaning of each field.

For example, customers who have not churned may not have a Churn_Date.

◆ Duplicate Records

Duplicate records were checked to prevent incorrect customer counts and KPI calculations.

◆ Data Types

Data types were standardized.

Examples:

Customer_ID → Text

Revenue → Decimal

Login_Count_30D → Whole Number

Sessions_30D → Whole Number

Last_Active_Date → Date

Churn_Date → Date

◆ Category Standardization

Customer segments, regions, statuses, and other categorical values were standardized.

◆ Invalid Values

Incorrect or inconsistent values were identified and corrected where appropriate.

◆ **Derived Fields**

Additional analytical fields were created, including:

- Days Since Last Activity
- Usage Decline Percentage
- Tenure Band
- Risk Score
- Risk Band
- Churn Status

◆ **Validation**

After cleaning, the dataset was checked for:

- ✓ Missing values
- ✓ Duplicate records
- ✓ Incorrect data types
- ✓ Invalid values
- ✓ Inconsistent categories

This prepared the dataset for SQL analysis and Power BI visualization.

7. SQL METHODOLOGY

SQL was used to perform structured data processing and analysis.

The SQL methodology was divided into several stages.

1. Table Creation

Database tables were created for customer-related information.

Customers

Subscription

Engagement

Revenue

Churn

Risk

Retention

2. Data Integration

SQL joins were used to combine related customer information using : Customer_ID

3. Data Transformation

SQL was used to create and transform analytical fields.

Examples include:

- Churn status
- Customer activity
- Usage decline
- Customer tenure
- Risk indicators

4. KPI Analysis

SQL queries were developed to calculate:

- Total Customers
- Active Customers
- Churned Customers
- Churn Rate
- Revenue and its Impact
- Customer-level metrics

5. Churn Signal Analysis

Customer behavior was analyzed using signals such as:

-  Usage decline
-  Reduced login activity
-  Reduced sessions
-  Long inactivity

6. Risk Scoring

Multiple customer signals were combined according to the project's predefined scoring rules.

Customers were then assigned a risk score and risk band.

7. Retention Analysis

SQL was also used to analyze:

- At-risk customers
- Recommended actions
- Assigned owners
- SLA requirements
- Intervention status and outcomes

8. CHURN DEFINITIONS

Clear definitions were used throughout the ChurnWatch project.

Churned Customer

A customer is classified as **churned** when their customer status satisfies the project's defined churn criteria.

Active Customer

A customer who is currently active and is not classified as churned.

Churn Rate

The basic churn rate is calculated as:

$$\text{Churn Rate} = \frac{\text{Churned Customers}}{\text{Total Customers}} \times 100$$

For example:

Churned Customers = 83

Total Customers = 1,000

$$\begin{aligned}\text{Churn Rate} &= 83 / 1,000 \times 100 \\ &= 8.3\%\end{aligned}$$

At-Risk Customer

An at-risk customer is an active customer showing one or more indicators associated with potential churn.

Examples:

 Declining usage

 Reduced login activity

 Reduced sessions

 Extended inactivity

The risk score helps organize these signals into a structured customer-risk view.

9. ⚡ RISK SCORE METHODOLOGY

The ChurnWatch risk-scoring framework is designed to identify active customers who may require retention attention.

The score considers multiple customer behavior and engagement indicators.

Main Signals

Signal	Purpose
🔒 Login Activity	Measures recent customer login behavior
📅 Sessions	Measures customer session activity
📉 Usage Decline	Identifies reduction in usage
⌚ Days Since Last Activity	Measures customer inactivity
🧩 Feature Usage	Measures product feature engagement

These signals are evaluated according to the project's predefined scoring rules.

The resulting score is converted into a **Risk Band**.

Example structure:

● Low Risk

● High Risk

● Medium Risk

● Critical Risk

A higher risk score represents a greater concentration or severity of the defined churn-risk signals.

The risk analysis is then connected with:

💰 Expected Revenue Impact

🕒 SLA Hours

🎯 Recommended Action

📅 Intervention Date

👤 Owner

📌 Intervention Status

This allows the project to move from **risk identification to retention operations**.

10. KPI DEFINITIONS

The following KPIs are used in the ChurnWatch dashboard.

 KPI	Definition
 Total Customers	Total number of customers in the dataset
 Active Customers	Customers currently classified as active
 Churned Customers	Customers classified as churned
 Churn Rate	Percentage of customers who have churned
 Total Revenue	Revenue associated with the analyzed customer population
 Revenue at Risk	Expected revenue impact associated with customers identified as at risk
 Average Revenue per Customer	Average revenue generated per customer
 High/Critical Risk Customers	Active customers in higher risk bands
 Usage Decline %	Percentage reduction in customer usage
 Days Since Last Activity	Number of days since the customer's latest activity

These KPIs provide a consolidated view of **customer health, churn, risk, and revenue exposure**.

11. POWER BI DASHBOARD

Power BI is used to convert the processed data into interactive analytical dashboards.

Microsoft's Power BI documentation describes Power BI as a platform for connecting to data, modeling and analyzing it, creating reports, and exploring insights.

PAGE 1 — EXECUTIVE DASHBOARD

The Executive Dashboard provides a high-level overview of customer performance.

KPI Cards

-  Total Customers
-  Active Customers
-  Churned Customers
-  Churn Rate
-  Revenue
-  Revenue at Risk

Purpose

This page helps users quickly understand the overall customer and churn situation.

PAGE 2 — CHURN ANALYSIS

The Churn Analysis page focuses on understanding historical churn patterns.

The analysis can be performed across:

- Customer Segment
- Tenure Band
- Region
- Subscription attributes
- Other available customer dimensions

Purpose

The page helps users identify patterns in customer churn and compare churn across different customer groups.

⚡ PAGE 3 — CUSTOMER RISK DASHBOARD

The Risk Dashboard focuses on active customers who may require attention.

It includes:

- ⚡ Risk Score
- Risk Band
- 💰 Expected Revenue Impact
- 📉 Usage Decline
- 🔑 Login Activity
- 📊 Sessions
- ⌚ Days Since Last Activity

Purpose

The page helps identify and analyze customers based on their risk level.

🎯 PAGE 4 — RETENTION OPERATIONS

The Retention Operations page connects customer risk analysis with operational actions.

The table contains:

Field	Purpose
Customer_ID	Identifies customer
Risk_Band	Shows risk level
Recommended_Action	Suggested retention action
Owner	Person/team responsible
SLA_Hours	Target response time
Intervention_Date	Date of intervention
Intervention_Status	Current intervention status

A filter can be applied to display:

Intervention_Status = Pending

This provides a practical view of retention activities that require attention.

12. 💡 KEY INSIGHTS

The ChurnWatch analysis provides insights across several areas.

👥 Customer Churn

The dashboard provides visibility into the total number of active and churned customers and the overall churn rate.

📊 Customer Engagement

Login count, session activity, feature usage, and inactivity provide useful indicators of customer engagement.

Customers showing declining engagement can be investigated further.

⌚ Customer Tenure

Customers can be analyzed across tenure groups:

0–6 Months

6–12 Months

12–24 Months

24+ Months

This helps identify differences in customer behavior across the customer lifecycle.

⚡ Customer Risk

Risk scoring allows active customers to be grouped into different risk bands.

This provides a structured way to identify customers requiring additional attention.

💰 Revenue Exposure

Expected revenue impact provides additional context by connecting customer risk with potential financial exposure.

🎯 Retention Operations

The retention page provides visibility into pending interventions, responsible owners, SLA requirements, and intervention outcomes.

13. RECOMMENDATIONS

Based on the analytical framework, the following retention practices can be considered.

1. Monitor High-Risk Customers

Customers in higher risk bands should be monitored more closely.

2. Investigate Declining Engagement

Customers showing reduced login, session, or feature usage can be investigated to understand the reason for declining engagement.

3. Consider Revenue Impact

Retention teams can consider both customer risk and expected revenue impact when prioritizing their work.

4. Use Customer-Specific Actions

Recommended actions should be connected to the customer's observed behavior.

5. Monitor SLA

Retention teams can track SLA hours to ensure interventions are addressed within the defined timeframe.

6. Track Intervention Status

Retention activities should be tracked using the project's defined intervention statuses.

7. Monitor KPIs Regularly

The Power BI dashboard can be refreshed regularly to monitor changes in:

- Churn
- Customer risk
- Revenue exposure
- Engagement
- Retention activities

14. ⚠️ LIMITATIONS

Although ChurnWatch provides a structured analytical framework, it has several limitations.

1. Dataset Size

The current dataset contains approximately 1,000 customers and may not represent the behavior of a much larger real-world population.

2. Data Availability

The accuracy of the analysis depends on the completeness and quality of available customer information.

3. Risk Score Assumptions

The risk score is based on predefined rules and available customer signals. It should be treated as an analytical indicator rather than a guaranteed prediction of future churn.

4. External Factors

Factors such as:

- Competitor activity
- Pricing changes
- Customer satisfaction
- Market conditions
- Economic conditions

may affect churn but may not be included in the current dataset.

5. Future Behavior

Customer behavior can change over time, so risk indicators and revenue-impact estimates may change after new data becomes available.

15. CONCLUSION

ChurnWatch demonstrates how customer data can be transformed into a structured **Customer Churn & Retention Analytics** solution using **Excel, SQL, Power Query, DAX, and Power BI**.

The project integrates customer information, engagement behavior, subscription information, revenue, churn data, risk scoring, and retention operations into a unified analytical workflow.

SQL provides the foundation for data integration, transformation, KPI calculations, churn-signal analysis, and customer-level risk analysis.

Power BI converts the processed information into interactive dashboards covering:

-  Executive Overview
-  Churn Analysis
-  Customer Risk
-  Retention Operations

A major strength of the project is that it goes beyond simply displaying historical churn numbers. It also considers **customer engagement signals, risk bands, expected revenue impact, recommended actions, owners, SLA hours, and intervention status**.

The project therefore demonstrates practical skills in:

 **Excel** →  **Data Cleaning** →  **SQL** →  **DAX** →  **Power BI** →  **Business Insights**

Overall, ChurnWatch provides a complete portfolio-level example of applying data analytics and business intelligence techniques to a customer retention problem.